

Previous: Harness-Bound Training

Single Harness



Claude Code



Bash



Read



Edit



Plan

Switch Harness



Openhands-SDK



Terminal



file_editor



task_tracker



Codex



exec command



exec command



update_plan



Ours: Hint—Harness Invariant Training

Real Trajectory



Code task



Terminal



Read



edit



Search



Plan

Harness Protocol



run_shell



read_file



edit_file



Search



track

same semantic trajectory

Counterfactual Harnesses



Vocabulary

bash

execute_bash

terminal



Granularity

split

unified editor

shell-only



Schema & Docs



Observations

plain

line-numbered

truncated



Completion

finish tool

final reply

Harness Invariant Agent



claude code

Codex

OpenCode

Openhands